

Cover Page

STAT 440

Project 2 Report

Instructor: Lloyd Elliott

Section Number: D100

Student Name: Stuart Siu, Elysa Lin, Geoffrey Ze-Yu Gao, Varinder Singh, Min Kaung Khant

Student Number: 301243574, 301556566, 301289048, 301575256, 301560051

Date: November 28, 2025

Question 1

Question 1 - Stuart

What techniques did you use to preprocess the data and visualize the data? What interesting conclusions could you make about the data based on the visualizations or summary statistics?

Preprocessing and Visualization Techniques

This project classifies kiln types from 724 satellite RGB images (256×256), each assumed to contain at least one kiln. Because a kiln occupies only ~5% of an image and appears at random locations, the model cannot rely on fixed spatial patterns. We therefore designed a preprocessing pipeline that automatically locates the most informative region before classification.

Redness Map

Kilns show a strong brick-red tone in most conditions. We created a simple “redness” map:

$$\text{heat} = r - 0.333 * g - 0.333 * b$$

This highlights kiln-coloured areas and suppresses vegetation and water, while preserving whites for B&W images. We inspected images at random and confirmed that kilns typically appear as bright, localized peaks in this map.

Sobel Edge Map

Colour alone often produced false positives from fields, soil or roads. To incorporate structural information, we converted RGB images to grayscale using the standard ITU-R BT.601 luminance weights¹ and draw a mask using Sobel edge technique². From our random inspection, kilns consistently generated sharp, coherent edges, while non-kiln regions produced weaker, noisier gradients.

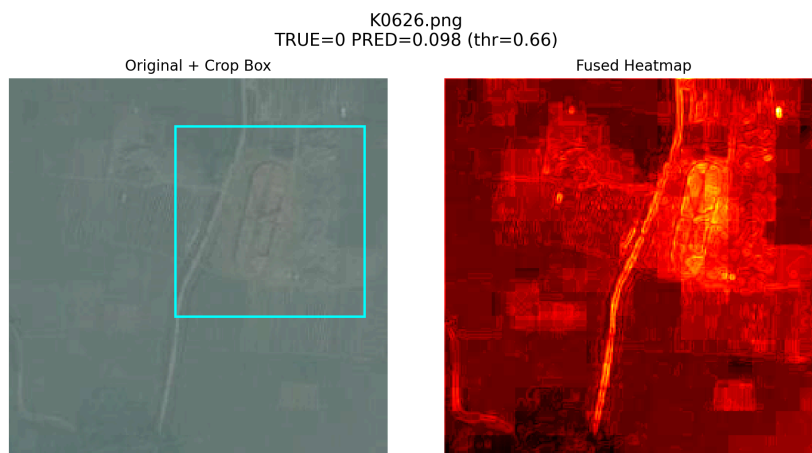
Final Formula

$$\text{fused} = (1 - w_{\text{edge}}) * \text{rgb_h} + w_{\text{edge}} * \text{rgb_h} * \text{sobel}$$

After visualizing the heatmap, we found tuning the weight w_{edge} to 0.6 is a sweet spot where kilns “shine” strongly even in degraded images, while most background objects are heavily suppressed. We then scanned the fused map with a Gaussian-weighted window and selected the highest-scoring 128×128 crop. We also quickly reviewed the correct and misclassified images after the predictions, and noted which crops were wrong from the start, knowing improvements were needed to the model.

Preprocessing versus Pretrained Features

We keep the final crop in true RGB so it matches the requirement from the pretrained ImageNet database. By letting our preprocessing pipeline handle kiln localization, the pretrained model no longer needs to start from scratch, and it can focus entirely on distinguishing kiln types.



3

¹ International Telecommunication Union, *Recommendation ITU-R BT.601-7: Studio encoding parameters of digital television for standard 4:3 and wide-screen 16:9 aspect ratios*, 2011.

https://www.itu.int/dms_pubrec/itu-r/rec/bt/r-rec-bt.601-7-201103-!!!pdf-e.pdf

² Gonzalez, R. C., & Woods, R. E. (2002). *Digital Image Processing* (2nd ed.). Prentice Hall.

³ This is an image selected randomly from the validation set, which is correctly classified as a fixed chimney kiln.

Question 3

Question 3 - Elysa

What models did you use to make your predictions, and what was the overall pipeline you developed to make your predictions? Which models worked best?

A series of models was developed and refined to generate predictions for the kiln classification task.

Baseline convolutional neural network

A baseline convolutional neural network was first trained on resized 256×256 RGB tiles in PyTorch. Several convolutional blocks with batch normalization and ReLU activations were followed by fully connected layers and optimized using binary cross entropy with the Adam optimizer. This baseline achieved a leaderboard score of 20.5%, so learning directly from full tiles without any kiln localization or transfer learning was found to be ineffective.

Heatmap guided EfficientNetB0 model

To improve performance, a fused heatmap guided cropping pipeline with EfficientNetB0 was introduced. For each image, a brick color map and a Sobel edge map were combined into a fused heatmap, and a Gaussian weighted sliding window was used to select a 128×128 crop centred on the most kiln-like region. These crops were passed into EfficientNetB0 without its original classification head, followed by a small dense layer for binary output. This model achieved a leaderboard score of 48.9%, representing an absolute gain of 28.4 percentage points over the baseline, so kiln focused cropping together with transfer learning provided a substantial and practically important improvement.

Final EfficientNetB1 ensemble with fused features

The final prediction pipeline was based on a more expressive multi input architecture with an EfficientNetB1 backbone. For each image, the fused heatmap was computed, the best 128×128 window was selected using the Gaussian weighted scoring scheme, and three global shape features were extracted from the fused map. The network then received three inputs for every kiln tile, namely the RGB crop processed by EfficientNetB1, a small convolutional branch applied to the fused heatmap crop, and the three dimensional shape feature vector, which were fused in a joint dense head. Training was carried out with strong regularization that included spatial jitter, random flips, mild rotations, zoom and contrast changes, as well as mixup applied to both images and features. A stratified ten fold cross validation procedure was used, and for each fold an independent model was trained and then applied with test time augmentation based on multiple jittered crops, with predictions averaged across augmentations and across folds. This final ensemble achieved a leaderboard score of 99.93%, which represented a very large improvement over both the baseline CNN and the single EfficientNetB0 model.

Overall comparison and best model

Overall, a consistent pattern was observed. Models that operated on full tiles without localization or shape information performed poorly, while each successive refinement that added fused heatmap cropping, transfer learning, shape features, an auxiliary fused heatmap branch, strong data augmentation, mixup, and cross validated ensembling produced clear gains in predictive performance. The final EfficientNetB1 ensemble with fused RGB and heatmap features, explicit shape descriptors, and test time augmentation achieved the highest leaderboard score and was therefore selected as the model used to generate the submitted predictions.

Question 5

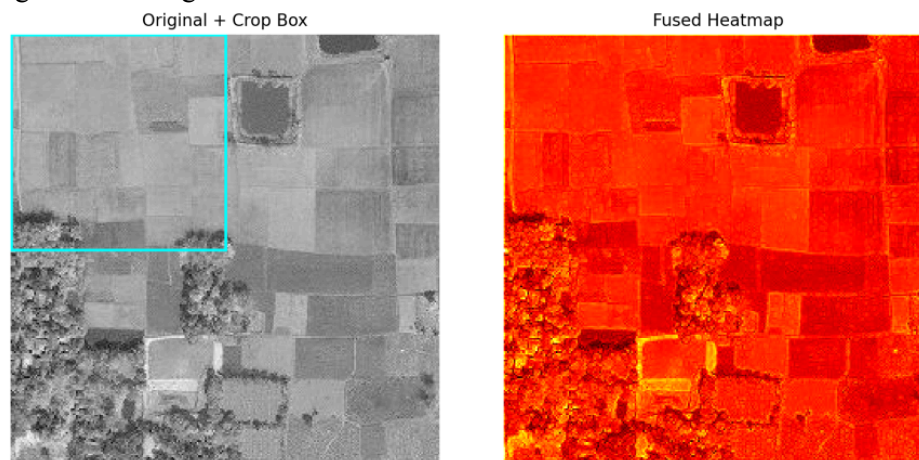
Question 5: Geoffrey

What technical or modelling challenges did you face during this project? How did you overcome or attempt to overcome those challenges?

Locating the Kiln in the Image

Identifying kilns in satellite images posed significant challenges due to varying image quality, with some in color and others black and white, and the difficulty of distinguishing small or poorly visible kilns against complex backgrounds such as dense vegetation. The variations in resolution and background clutter made it difficult to accurately locate kilns, especially in lower-quality images.

To address this, we used a combination of image processing techniques. RGB heatmaps were used to highlight areas likely containing kilns, while edge detection maps helped outline structures and potential kiln locations. Shape-based features, such as aspect ratios and contours, were also integrated to refine the detection process. These methods allowed us to improve the model's ability to identify kilns, even in images with significant background noise or low resolution.



Model Training and Computational Constraints

Training the model on a dataset of over 1,600 satellite images was computationally intensive, primarily due to the use of convolutional neural networks (CNNs) in our model. This requires significant resources each epoch due to the large number of parameters. With each epoch requiring substantial computational resources, the long runtimes of multiple training folds for cross-validation, along with up to 150 epochs, further compounded the challenge. As a result, the training process takes several hours.

To optimize this process, we implemented early stopping to monitor the validation Area Under the Curve (AUC). If the AUC did not improve for 20 consecutive epochs, training was halted early, preventing unnecessary computational waste. Additionally, by using the best-performing model weights based on the highest validation AUC, we reduced overfitting and improved overall efficiency. To further manage the long training times, we set the model to train overnight, ensuring continuous progress without needing to monitor it constantly. These optimizations together helped minimize computational costs while maintaining performance.

Question 6: Varinder

Question 6: Varinder

If you were to continue working on this project, what are some more techniques or approaches that you would try? Why do you think those approaches would work?

To further improve the fused-heatmap plus crop-based classifier, several promising approaches can meaningfully enhance accuracy and robustness. Below are four techniques that can be used to further strengthen the model:

1. Domain-Specific Self-Supervised Pretraining

ImageNet consists mostly of eye-level consumer photos, which is not ideal for kiln imagery. Better feature initialization can be achieved by performing self-supervised pretraining directly on the unlabeled kiln dataset (for example SimCLR, BYOL, or MoCo) or by using weights trained on perspectives that are more similar to ours, such as satellite or aerial imagery datasets like SATIN or GeoNet. Both options provide more appropriate low-level texture and structural features and typically lead to a measurable AUC improvement.

2. Multi-Crop or Multi-Region Modeling

The current pipeline uses only the single highest-scoring 128×128 crop. Important cues may appear in several locations. Extracting multiple high-score fused-heatmap regions and applying multiple-instance learning (MIL) or transformer-based pooling allows the model to integrate more spatial evidence. This reduces the chance of missing subtle defects and improves overall robustness.

3. Stronger Backbones (ConvNeXt, EfficientNetV2, ViT)

Modern architectures such as ConvNeXt-T or ViT-Small provide more powerful texture and global-context representations than EfficientNetB0. Replacing the backbone with one of these tends to increase discriminative power and reduce overfitting, leading directly to higher validation AUC.

4. Learned Heatmap Generation (Small UNet)

Instead of using fused maps based on fixed rules such as RGB differences and Sobel edges, a small UNet can be trained to generate a learned defect-likelihood heatmap. This allows the crop-selection step to adapt during training and capture more complex visual patterns. Learned heatmaps are generally more stable and informative than manually constructed ones.

These four techniques address the main challenges of the dataset including domain shift, spatial variability, limited labels, and texture complexity. Each approach is supported by results from industrial vision tasks and is likely to improve model accuracy and generalization.

Question 7 :)

Question 7 - Min

How did you organize work on this project as a team? How did you organize sharing code and data? What worked well, and how could you improve your organization style?

For this project, our team refined our workflow by improving how we shared code, managed tasks, and communicated progress. We have incorporated much of what we have learned from Project 1. We continued using GitHub and Discord as our primary collaboration tools, but we incorporated more structured practices that made our work more organized.

How do we organize code and data?

We maintained a branch-based development workflow in GitHub, but with more descriptive naming conventions and comments for better code readability. We made the naming conventions such that it is immediately clear what each branch contained and who was responsible for it.

We also focused more on improving the overall readability and maintainability of our code. We did it by intentionally separating the code to different sections, such as reproducibility setup (random seeds, package loading), feature engineering section, etc. This approach made it much easier for team members to understand, modify, and build upon one another's work. This structure also helped reduce merge conflicts and kept our experimentation more organized, as we added new features. Once a branch reached the highest prediction performance, the update was merged into main, and it became the new baseline branch.

How do we communicate?

We continued to use Discord as our primary communication channel. We held weekly meetings so that our team members can report which features they are working on, the different parameters they are testing the model with, and different ideas for how to approach the project.

What worked well?

Overall, our project logistics became more structured, largely because the team had already developed familiarity with the tools and with each other during the previous project. This allowed us to communicate and work on the project more efficiently.

Areas for Further Improvement

Although the project was relatively small and our prior experience working together made coordination easier, we still could have benefited from adopting a more formal task-tracking system. We relied primarily on Discord for delegating tasks and communicating who was working on which part of the pipeline. While this approach was sufficient for a fast-paced project with five team members, using GitHub Issues would have provided clearer documentation and reduced the chances of overlapping work.